

Zehao Liu

Data Engineer • Machine Learning Engineer • Artificial Intelligence Engineer • Data Scientist • Data Analyst
zliu5660@gmail.com | GitHub | LinkedIn | Sydney, Australia

Professional Summary

Early-career technical professional with a CS and mathematics foundation, primarily targeting Data Engineer roles and also open to ML, AI, and analytics positions. I combine fast learning, structured decomposition, and a product-aware delivery mindset to build reliable pipelines and analysis workflows.

- Target roles: Data Engineer (primary); also open to ML Engineer, AI Engineer, Data Scientist, and Data Analyst.
- Working pattern: learn quickly, structure the work, ship, and review with data and feedback.
- Portfolio evidence: pipelines, model evaluation, dashboards, and reproducible delivery.

Professional Strengths

- Connect engineering work to business goals and user experience across the full delivery loop.
- Break complex tasks into structured plans with clear priorities and risk control.
- Use AI-assisted coding, research, and documentation with verification and explainability as guardrails.
- Support cross-functional teams through Jira, documentation, reporting, and client communication.
- Stay steady under pressure while balancing detail quality and response speed.

Education

The University of Sydney

Master of Computer Science specialized in Data Science and AI

- Data pipelines, data engineering, NLP, machine learning, and data analysis / modeling.

Sydney, Australia

Feb 2025 - Aug 2026

Wilfrid Laurier University

Honours Bachelor of Computer Science and Mathematics

- Algorithms, proofs, statistics, databases, AI, and network security.

Waterloo, Canada

Sep 2019 - Jun 2024

Technical Focus

- Data engineering, SQL, databases, ETL, and reproducible pipelines.
- Machine learning, model evaluation, deployment, and reproducible ML workflows.
- Data analysis, dashboarding, reporting, and stakeholder-ready insight.
- AI-assisted development workflows with verification, documentation, and explainability guardrails.

Skills

Technologies: caret, CatBoost, Docker, ggplot2, GitHub Actions, JavaScript, Konva.js, LightGBM, Matplotlib, MQTT, Node.js, NumPy, Pandas, Python, R, Render, Scikit-learn, Streamlit, TensorFlow / Keras, Vercel, Vite, WebSocket, XGBoost

Projects

NEM Monitoring Dashboard | [GitHub](#) | [Demo](#)

Data Engineering

- I designed and implemented the ingestion, publishing, and dashboard layers, then documented the deployment and fallback behavior.
- The dashboard presents live facility updates with stable local and Render deployment paths, plus a fallback replay mode when the stream is stale or unavailable.

Infinite Canvas Studio | [GitHub](#) | [Demo](#)

Software Engineering

- I contributed across product scoping, frontend canvas implementation, Edit/Present mode workflows, WebSocket collaboration, documentation, testing, and deployment.
- The app supports temporary online rooms for collaboration, separate frontend and relay deployments, and export paths for JSON, single-file HTML, and PROJ folder formats.

Medical Image ML Algorithm | [GitHub](#)

AI / ML

- I designed the end-to-end pipeline, implemented the CLI and model training code, added Grad-CAM and error analysis, and documented reproducible setup for local and Apple Silicon runs.

- The CNN reached 90.1% test accuracy, outperforming PCA + Random Forest (65.7%) and PCA + MLP (69.3%). Grad-CAM and confusion-matrix analysis highlighted where spatial features helped and which tissue classes were most often confused.

EduAttain Prediction | *GitHub*

Data Analytics

- I contributed to data ingestion and cleaning against ABS metadata, exploratory analysis, model benchmarking, and the Python automation for XGBoost, LightGBM, and CatBoost tuning.
- The workflow produced a cleaned modeling dataset, exploratory reports, and tuned XGBoost, LightGBM, and CatBoost pipelines alongside R baselines such as Random Forest and SVM for highest-attainment classification.